

AI Heart Health Predictor

Market Research Report & MVP Proposal

Submitted by: Honey Jaiswal | April 2026 | Target Launch: May 2026

1. Idea Summary

Problem Statement	Many young adults (ages 18–40) lack easy, accessible tools to predict their heart disease risk and take preventive action early.
Target Users	Busy professionals and young adults aged 18–40 who are health-conscious but time-constrained.
Proposed Product	An AI-powered Heart Health Predictor web app — users enter health metrics and lifestyle data, the AI predicts their risk level, and they receive personalised preventive advice.
Unique Value	Combines prediction + personalised recommendations + privacy in one lightweight, accessible platform — unlike existing fragmented solutions.
MVP Deadline	1 May 2026

2. Market Research — Existing AI Applications

The following applications represent the current competitive landscape for AI-driven heart health and cardiovascular risk tools. Each addresses part of the problem but leaves a meaningful gap your idea can exploit.

Apple Health / Fitbit (Wearables)	Fitness & Biometric Tracking
Problem Solved: Tracks real-time biometrics — steps, resting heart rate, SpO2, sleep patterns — to promote daily activity and flag anomalies through on-device algorithms. Shortcoming: Requires expensive hardware (Apple Watch, Fitbit). Metrics are observational rather than predictive. Does not model individual cardiovascular risk or provide clinical-grade guidance. No actionable lifestyle coaching based on risk score.	

Antshrike (Before Health Intelligence)	Clinical CVD Event Prediction
Problem Solved: Uses AI on Apple Watch biometric time-series data to predict the timing of imminent major cardiovascular events (heart attack, stroke) with high accuracy for at-risk patients. Shortcoming: Requires Apple hardware; targets already-diagnosed or clinically at-risk patients rather than healthy young adults seeking prevention. Highly clinical, inaccessible to the average 18–40-year-old. Not yet widely available (freemium U.S. launch targeted for 2025).	
AHA / WHO Online Risk Calculators (e.g., ASCVD Risk Estimator)	Static Risk Scoring Tools
Problem Solved: Provides a one-time cardiovascular disease risk score based on age, cholesterol, blood pressure, smoking status, and diabetes using validated clinical formulas (Framingham, PCE). Shortcoming: Not AI-driven; relies on fixed statistical formulas with no personalisation over time. UI is clinical and intimidating for non-medical users. Gives a number with no follow-up guidance, habit tracking, or motivational nudges. No user account or history.	
Welltory	Stress & HRV-Based Wellness
Problem Solved: Analyses heart rate variability (HRV) from smartphone cameras or wearables to deliver personalised energy, stress, and recovery scores, and suggests lifestyle adjustments. Shortcoming: Focuses on stress and recovery rather than heart disease risk prediction. Does not incorporate cholesterol, blood pressure, family history, or clinical risk factors. Wellness-oriented, not medically credible for CVD prevention among young professionals.	
Idoven / CardioAI (B2B Platforms)	Clinical ECG Analysis (B2B)
Problem Solved: AI platforms that analyse ECG data to detect arrhythmias, predict heart failure, and support clinical diagnosis — used by hospitals and employers for large-scale cardiac screening. Shortcoming: Designed for healthcare providers and enterprises, not individual consumers. Requires ECG hardware or EHR integration. Not accessible as a self-service web app for a young adult wanting a quick personal risk check. High regulatory and cost barriers for end users.	
General LLM Chatbots (ChatGPT, Gemini Health Q&A;)	General-Purpose Health Q&A;
Problem Solved: Answers health-related questions conversationally, explains symptoms, and provides general lifestyle advice based on user descriptions. Shortcoming: Not specialised in cardiac risk assessment. Does not collect structured health data or generate a calibrated, longitudinal risk score. Lacks memory between sessions for tracking improvement. Provides generic disclaimers rather than actionable, personalised guidance.	

3. Identified Gap

There is no simple, consumer-friendly web app that lets a young adult (18–40) input their basic health metrics and lifestyle habits — without requiring hardware, clinical data, or a doctor — and instantly receive (a) an easy-to-understand personalised heart disease risk score powered by AI, and (b) specific, actionable lifestyle recommendations tied directly to that score. Existing tools are either hardware-dependent, clinically complex, hardware-locked, static calculators, or generic chatbots. The intersection of accessible input + AI risk prediction + personalised coaching in a single lightweight web experience is wide open.

Gap Dimension	What Exists	What Is Missing
Accessibility	Wearables / clinical portals	Zero-hardware web app
Risk Prediction	Static formulas or ECG-based AI	LLM-driven personalised risk score
Actionable Output	Score only, or generic tips	Score + specific lifestyle changes
Target Audience	40+ patients or enterprises	Healthy young adults (18–40)
Privacy	Data sold or wearable-synced	User-controlled, encrypted inputs

4. MVP Proposal — CardioCheck AI

Product Name: CardioCheck AI — "Know your heart risk in 2 minutes."

CardioCheck AI is a focused, single-page web app. A user fills in a short health form (age, weight, blood pressure, cholesterol level, smoking habits, physical activity, family history), submits it, and immediately receives: a colour-coded risk score (Low / Moderate / High) explained in plain language, plus three to five specific, personalised lifestyle changes they can start today. No login required for the first use. No wearable. No clinical jargon. One clear outcome per session.

Core User Flow

Step	Action	Detail
1	Open the web app	Single-page form — no registration needed
2	Enter health metrics	Age, gender, weight/height, BP, cholesterol, smoking, exercise frequency, family history
3	Click 'Check My Risk'	Form data sent to Node.js backend
4	AI analyses data	Gemini API generates risk assessment + advice
5	View results	Colour-coded risk card + 3–5 actionable tips
6	Optional: Save / Share	User can save PDF or share link (Supabase if enabled)

Tech Stack (Minimal & Free-Tier Friendly)

Frontend — Next.js (React): Single-page app, Tailwind CSS for styling. Deployed free on Vercel. Fast build, easy iteration.
Backend — Node.js + Express: Lightweight REST API. Single endpoint: POST /assess. Deployed free on Render.
AI / LLM — Google Gemini API: Gemini 1.5 Flash — fast, generous free tier. Structured prompt returns risk level + tips as JSON.
Database — Supabase (optional): Only needed if user wants to save history. Free tier is sufficient for MVP validation.
Auth — None (MVP phase): No login required in v1. Reduces friction, speeds validation.
Hosting — Vercel + Render: Both offer free tiers. Frontend on Vercel, API on Render. Zero cost to launch.

Gemini Prompt Architecture (Core Logic)

The backend sends a structured system + user prompt to Gemini. The system prompt instructs the model to act as a preventive cardiology assistant, respond only in JSON, and never give a clinical diagnosis. The user prompt passes the form data as key-value pairs. Gemini returns a JSON object: { risk_level, risk_score_pct, summary, recommendations[] }. The frontend renders this directly — no parsing ambiguity.

System: You are a preventive cardiology risk advisor. Based on the health inputs provided, return ONLY a valid JSON object with these keys: risk_level (Low/Moderate/High), risk_score_pct (0-100 integer), summary (2 sentences, plain English), recommendations (array of 4 strings). Do NOT provide a clinical diagnosis. Do NOT add any text outside the JSON.

MVP Build Timeline (2–3 Weeks)

Week	Focus	Deliverables
Week 1	Foundation	Next.js project setup, Tailwind styling, health form UI, Node.js Express API scaffold, Gemini integration
Week 2	Core Feature	End-to-end flow working locally: form → API → Gemini → risk card. Error handling, loading states
Week 3	Deploy & Validate	Deploy frontend to Vercel, API to Render. Share with 10–20 target users. Collect feedback

What to Deliberately Exclude from v1 (Avoid Feature Bloat)

Feature to Skip	Reason
User accounts / authentication	Adds 3–5 days of build time and friction for new users. Add only if retention data justifies it.
Wearable / Apple Health integration	Hardware dependency kills accessibility — the core differentiator is zero-hardware.

Historical tracking / dashboard	Nice to have, not needed to validate the core value proposition.
Payment / premium tier	Monetisation comes after proving people return and value the product.
Mobile native app (iOS/Android)	A responsive Next.js web app covers 95% of mobile use cases. Ship the web version first.

5. MVP Validation Metrics

These are the minimal signals to look for in the first two weeks after launch. If any of these are positive, double down. If none are, interview users before adding features.

Type	Metric	Description	Target
Primary	Completion rate	% of users who complete the form and view their results	> 60%
Primary	Return visits	Users who re-submit with updated metrics within 7 days	> 20%
Signal	Social share / copy link	Users who share their risk card	> 10%
Signal	Qualitative feedback	Users rating the advice as 'useful' or 'actionable'	> 70% positive
Guard	API error rate	Gemini call failures or malformed JSON responses	< 5%
Guard	Page load time	Time to interactive on 4G mobile	< 3 seconds

6. Conclusion

The market for AI-driven heart health tools is active and growing, but it skews clinical, hardware-dependent, and enterprise-focused. There is a clear, underserved space for a lightweight web app that helps a healthy young adult get a personalised, AI-driven heart disease risk score in under two minutes — with no hardware, no login, and no medical jargon.

CardioCheck AI addresses this gap with the simplest possible implementation: a Next.js form, a Node.js/Express endpoint, and a Gemini API call. The entire stack runs free on Vercel and Render, the build can be completed in three weeks, and the idea can be validated with real users before any complexity is added.

The guiding principle for the MVP is: **one form, one risk score, one set of personalised tips**. Everything else is noise until users prove they find value in that core loop.